



Synthesis and antifouling properties of carbazole amide derivatives with fluorescent functional structure

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ABSTRACT

Tin-free self-polishing antifouling coatings are prevalent today. However, its high Cu_2O content endangers the marine environment. Leveraging the fluorescent properties of coating to prevent fouling is an environmentally friendly antifouling coating development. In this paper, four unreported fluorescent structural carbazole amide derivatives (FSCAD) were synthesized by Friedel–Crafts alkylation reaction. Their fluorescence, antibacterial, and algal inhibition properties were examined. The fluorescent properties test results showed that the four FSCAD exhibited high fluorescence properties. The results of antibacterial test and anti-algae growth test showed that the maximum bacterial inhibition rate can reach 90%, and the growth inhibition rate of three kinds of microalgae can reach more than 75%. It was added to the antifouling coating system as an antifouling agent instead of the traditional antifouling agent Cu_2O . The abrasion rate, contact angle, surface morphology, and roughness of the coatings were determined by dynamic simulation experiments, and the antifouling properties of the coatings were evaluated by the anti-algae sedimentation test and marine environment antifouling experiments. The results of dynamic simulation experiments showed that adding FSCAD does not affect the wettability and surface state of the coating. Anti-algae sedimentation test exhibited that marine antifouling coatings with FSCAD showed excellent inhibition on the sedimentation of microalgae. The anti-fouling performance of the coating was tested in the marine environment antifouling experiments. The results showed that the coating exhibited excellent anti-fouling performance, with minimal biofouling was attached to the surface of the coating. Especially, the antifouling performance with compound A was better than that of chlorothalonil.

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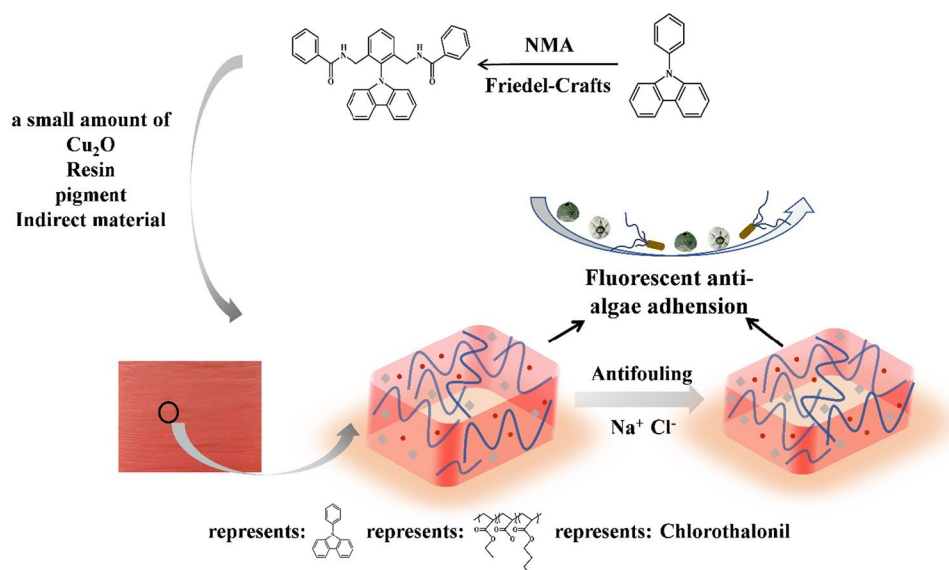
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GRAPHICAL ABSTRACT



Introduction

Marine biofouling refers to the attachment of various micro-organisms, animals, and plants in the ocean to human-made structures [1]. This phenomenon represents a significant long-term challenge for ships at sea, affecting not only their speed and maneuverability but also their durability [2, 3]. Consequently, the prevention of marine pollution is of great significance to the development and utilization of marine resources [4].

Antifouling coatings are a widely used method due to their simple operation and low cost [5]. Currently, the mainstream antifouling coatings are self-polishing antifouling coatings with zinc acrylate resin as film-forming material and Cu_2O as main antifouling agent [6–8]. However, with the increasing demand for marine ecological protection, traditional copper-based antifouling coatings are toxic due to Cu_2O . The biological concentration of Cu_2O in seawater results in the mutation of marine organisms' genes, which in turn affects human health via the food chain [9–12]. Consequently, many countries have included Cu_2O in the list of 'high-polluting substances' [13]. Therefore, improving the antifouling property of coatings has become the future direction of development for antifouling coatings.

Natural product antifouling agents are derived from marine and terrestrial organisms that exhibit antifouling activity. However, the yields of these

agents are limited, and the extraction process is challenging. Consequently, researchers have conducted research on the development of synthetic antifouling agents based on natural antifouling agents. Introducing the chemical structure with biocidal activity into the coating is an effective method to improve its antifouling properties. Wang et al. prepared three novel green antifouling coatings with high efficiency, containing phthalimide derivatives inspired by capsaicin (PDIC-AC). Antifouling tests showed that the PDIC-AC efficiently inhibited bacterial growth and diatom adhesion. Through the marine environment antifouling experiments, the PDIC-AC maintained a good antifouling effect for 9 months [14]. Wang et al. prepared a modified acrylic resin containing capsaicin derivatives by free radical condensation and polymerization. The marine environment antifouling experiments demonstrated that the resin exhibited an antifouling period of up to 22 months. However, the above antifouling active structure only achieves the anti-fouling effect through chemical activity and is only effective for a limited range of fouling organisms, this is insufficient to meet the requirements of multi-sea area and diversified fouling organisms [15].

In recent years, antifouling coatings with fluorescent structures have emerged as an innovative antifouling technology. Moreover, the fluorescence properties have been proven to effectively improve the antifouling ability of the coating. On the one hand,

fouling organisms such as barnacle larvae in marine environments are naturally equipped with the ability to evade luminescent objects. Therefore, when artificial facilities such as ships are coated with fluorescent coatings and emit fluorescence, these organisms will stay away from such facilities, thus achieving the purpose of anti-fouling [16]. On the other hand, fluorescence can interfere with the chloroplast spatial structure of microalgae, causing the microalgae to photosynthesize first instead of attaching and settling [17, 18]. Hao et al. introduced waterproof long after-glow phosphors (WLAP) into waterborne silicone elastomer by physical blending method. The composite coating stores energy during the day and emits various colors of light at night, which affects the regular physiological activities of marine bacteria. The combination of fouling-release and fluorescence properties in the WLAP/PDMS composite coating results in a synergistic antifouling effect, with a bacterial adhesion rate of only 3.7% [19]. Jin et al. inspired by fluorescent corals, prepared a PDMS-phosphor-silicone rubber sandwich structure coating that emits weak light during night hours. The coating is effective in inhibiting various algae and bacteria and exhibits broad-spectrum antifouling properties. The combination of luminescent material and antifouling coating provides a new idea for environmentally friendly antifouling coatings.

In this work, a series of fluorescent structural carbazole amide derivatives (FSCAD) were synthesized by Friedel–Crafts alkylation reaction. They were substituted for Cu_2O as antifouling agents in the antifouling coating system. A multifunctional antifouling coating with chemical activity, fluorescence, and self-polishing was prepared using chlorothalonil as a control. This coating provides an important basis for the development of environmentally friendly multifunctional marine antifouling coatings.

Experimental

Materials

N-phenyl-carbazole, N-ethyl carbazole, N-hydroxymethyl benzamide, N-hydroxymethylphthalimide phthalic anhydride and acrylic acid monomers were purchased from Shanghai Aladdin Biochemical Technology Co, and anhydrous ethanol, concentrated sulfuric acid were purchased from Sinopharm Chemical

Reagent Co. All reagents were used according to the instructions.

Synthesis and characterization of FSCAD

N-hydroxymethylbenzamide (0.12 mol) and N-phenylcarbazole (0.1 mol) were added to a 500 mL three-necked flask with an electric stirring device, and anhydrous ethanol was added to dissolve. After half an hour of reaction under electromagnetic stirring, an appropriate amount of concentrated sulfuric acid was added dropwise. The reaction was carried out at 40 °C for 72 h. At the end of the reaction, the white solid produced by the reaction was filtered, and the crude product was fully washed with deionized water to neutralize. The pure product A was obtained by recrystallization with anhydrous ethanol, dried, and sealed.

According to the synthesis method of the above compound A, the other three compounds were synthesized in this way, and their structures were characterized as follows:

N-(3-phenylcarbazole)-2,6-di-methylbenzamide (A): IR (KBr, pellet, cm^{-1}): 3305.88 ($\nu_{\text{N-H}}$), 1635.14 ($\nu_{\text{C=O}}$), 1536.86, 1501.24, 1488.42 ($\nu_{\text{C=C}}$), 1075.37 ($\nu_{\text{C-N}}$), 697 ($\gamma_{\text{C-H}}$); ^1H NMR (DMSO, TMS, 600 MHz) δ : 4.65 (d, 4H, $-\text{CH}_2-$), 7.34 (d, 2H, Ph-H), 7.43 (dd, 2H, Ph-H), 7.47 (t, 4H, Ph-H), 7.53 (dt, 4H, Ph-H), 7.60 (m, 2H, Ph-H), 7.63 (t, 2H, Ph-H), 7.93 (m, 4H, Ph-H), 9.18 (d, 2H, N-H).

N-phenylcarbazole-2,6-di-methylphthalimide (B): IR (KBr, pellet, cm^{-1}): 1713.47 ($\nu_{\text{C=O}}$), 1593.83, 1499.82, 1449.97 ($\nu_{\text{C=C}}$), 1232.05 ($\nu_{\text{C-N}}$), 749.20 ($\gamma_{\text{C-H}}$); ^1H NMR (DMSO, TMS, 600 MHz) δ : 4.94 (d, 4H, $-\text{CH}_2-$), 7.31 (d, 2H, Ph-H), 7.40 (dd, 2H, Ph-H), 7.51 (t, 1H, Ph-H), 7.57 (m, 2H, Ph-H), 7.65 (t, 2H, Ph-H), 7.89 (m, 8H, Ph-H), 8.22 (d, 2H, Ph-H).

N-ethylcarbazole-3-methylphthalimide (C): IR (KBr, pellet, cm^{-1}): 1713.47 ($\nu_{\text{C=O}}$), 1603.80, 1489.85, 1468.49 ($\nu_{\text{C=C}}$), 1236.62 ($\nu_{\text{C-N}}$), 750.62 ($\gamma_{\text{C-H}}$); ^1H NMR (DMSO, TMS, 600 MHz), 4.42 (m, 3H, $-\text{CH}_3$), 4.94 (s, 2H, $-\text{CH}_2-$), 7.18 (t, 1H, Ph-H), 7.44 (m, 2H, Ph-H), 7.58 (dd, 2H, Ph-H), 7.86 (m, 2H, Ph-H), 7.91 (m, 2H, Ph-H), 8.13 (m, 2H, Ph-H).

N-(3-ethylcarbazole)-3-methylbenzamide (D): IR (KBr, pellet, cm^{-1}): 1639.31 ($\nu_{\text{C=O}}$), 1529.64, 1454.15, 1484.06, ($\nu_{\text{C=C}}$), 1072.45 ($\nu_{\text{C-N}}$), 749.15 ($\gamma_{\text{C-H}}$); ^1H NMR (DMSO, TMS, 600 MHz), 1.30 (t, 3H, $-\text{CH}_3$), 4.43 (q, 2H, $-\text{CH}_2-$), 4.65 (d, 2H, $-\text{CH}_2-$), 7.18 (t, 2H, Ph-H), 7.47 (m, 4H, Ph-H), 7.53 (m, 1H, Ph-H), 7.58 (dd, 2H,

Ph-H), 7.95 (m, 2H, Ph-H), 8.11 (m, 2H, Ph-H), 9.19 (t, 1H, N-H). (IR, ^1H NMR of the four compounds are shown in Figs. S1–S8 of the Supplementary Material).

Quantum chemical calculation of FSCAD

The initial configuration of the molecule was obtained by the Chem3D program. Gaussian09W performed DFT-B3LYP/6-31G geometric optimization of the molecule. Density functional theory (DFT)-B3LYP was used to perform quantum chemical calculations on fluorescent compounds to obtain the fully optimized molecular geometry, bond length, bond angle, dihedral angle, electronic structure, and other properties of the compound [20, 21] (Tables S2–S5).

Fluorescent spectra analysis

A certain amount of sample was dissolved and the fluorescence emission spectrum was obtained by the fluorescence spectrophotometer. Scanning voltage 700 V, slit width EX/EM: 2.5 nm/2.5 nm.

Antibacterial property test

In this work, *Marine Pseudomonas* was selected as the test strain, and the dilution coating plate method was used to evaluate the antibacterial properties of FSCAD [22, 23]. Before the antimicrobial test, all laboratory supplies were sterilized in an autoclave. The concentration of bacterial solution was 10^8 CFU/mL, and a certain amount of 3 mg/mL compound acetone solution was added. The bacterial solution was gradually diluted by a tenfold dilution method. 200 μL of the diluted solution was evenly spread on the solid medium and incubated at 37 °C for 18 h. The formation of bacterial colonies was observed and the number of viable bacteria T_0 was calculated. The number of viable bacteria T_n was calculated and the antibacterial rate was calculated by the following formula:

$$T = \frac{T_0 - T_n}{T_0} \times 100\%$$

where T is the bacteriostatic rate and T_0 and T_n are the number of colonies on the control and experimental agar plates, respectively.

Algae inhibition property test

Phaeodactylum tricornutum and *Chlorella pyrenoidosa* were used as the tested algae species. The maximum absorption wavelength of two microalgae was scanned by an ultraviolet spectrophotometer. The absorbance was measured at the maximum absorption wavelength, and the concentration of microalgae was determined by blood cell counting plate. The standard working curve was drawn accordingly, in which the abscissa was absorbance and the ordinate was concentration.

Algae cultured to the exponential growth cycle was diluted to the appropriate concentration and mixed homogeneously in a conical flask with 100 mL of algal solution and 1 mL of FSCAD. The algae solution without the FSCAD was used as the blank control. The absorbance value of each algae solution was measured at the maximum absorption wavelength every 24 h. According to the linear relationship between the absorbance value and the concentration of the algae solution, the time-concentration curve of the algae solution was drawn. The effect of the FSCAD on the growth curve of the microalgae was determined [24, 25]. The artificial climate chamber was illuminated with a light intensity of 4000 lx, with a 14:10 ratio of light to dark, and maintained at a temperature of 21 °C. The seawater was filtered through a cellulose filter membrane and sterilized at 121 °C. The microalgae inhibition rate was calculated by the following formula:

$$N = \frac{N_0 - N_n}{N_0} \times 100\%$$

where N is the inhibition rate of microalgae, N_0 is the concentration of microalgae in the control group, and N_n is the concentration of microalgae in the experimental group. Each experiment makes three groups of parallel control, and the average was taken as the inhibition rate of microalgae.

Preparation of antifouling coatings

As shown in the supplementary materials (Table S1), The FSCAD was added to the antifouling coating in a doping manner to prepare a fluorescent antifouling coating.

Anti-algae sedimentation property test of fluorescent antifouling coatings

When the exponential growth period was reached, *Nitzschia closterium*, *Chlorella pyrenoidosa*, and *Phaeodactylum tricornutum* were diluted with seawater to a specific absorbance. The fluorescent antifouling coating sample was placed at the bottom of the culture dish, and 40 mL of algae solution was added. Two groups were cultured under different conditions. One is in natural conditions, and the other is in completely dark conditions. The algae were stirred at the same time every day. After 7 days, the antifouling coating sample was removed and the surface was rinsed with the algae solution to remove the unattached microalgae. Then, the coated samples were rinsed with 40 mL seawater into another glass culture dish. The absorbance of the rinse solution was measured by an ultraviolet spectrophotometer, and the concentration of the algae was calculated according to the standard working curve of algae growth.

Testing the effect of FSCAD on the morphology of diatom cells

Following a 7 days cultivation period of *Phaeodactylum tricornutum*, 5 mL of algal solution was removed to a centrifuge tube. Subsequently, the algal cell precipitate

was collected and the algal cells were fixed with glutaraldehyde solution at 4 °C. Phosphate buffer solution (PBS) was prepared and used to rinse the algal cells three times. The cells were then dehydrated using a gradient of ethanol solutions, with concentrations of 30%, 50%, 70%, 80%, 90% and 100%. After 9 dehydration was completed, the cells were freeze-dried, and their morphology was observed using scanning electron microscopy. [25, 26]

Self-polishing performance of antifouling coatings

As shown in Fig. 1, the self-polishing performance of the coating was tested by dynamic padding simulation experiments. The coating was painted on well-polished acrylonitrile/butadiene/styrene copolymer (ABS) plates (350 mm × 100 mm × 3 mm) (Fig. 1a). The thickness of the coating was adjusted to be ~ 300 µm by the thickness gauge. The dried ABS plates were placed in a dynamic padding simulation device equipped with seawater as shown in Fig. 1b and adhered to the inner wall of the device for dynamic padding simulation experiments. After 10 days, the ABS plate was washed with deionized water multiple times until the surface was free of contaminants and dried to constant weight. Seven days were a cycle, and a total of five cycles were tested. The dynamic contact angle, loss

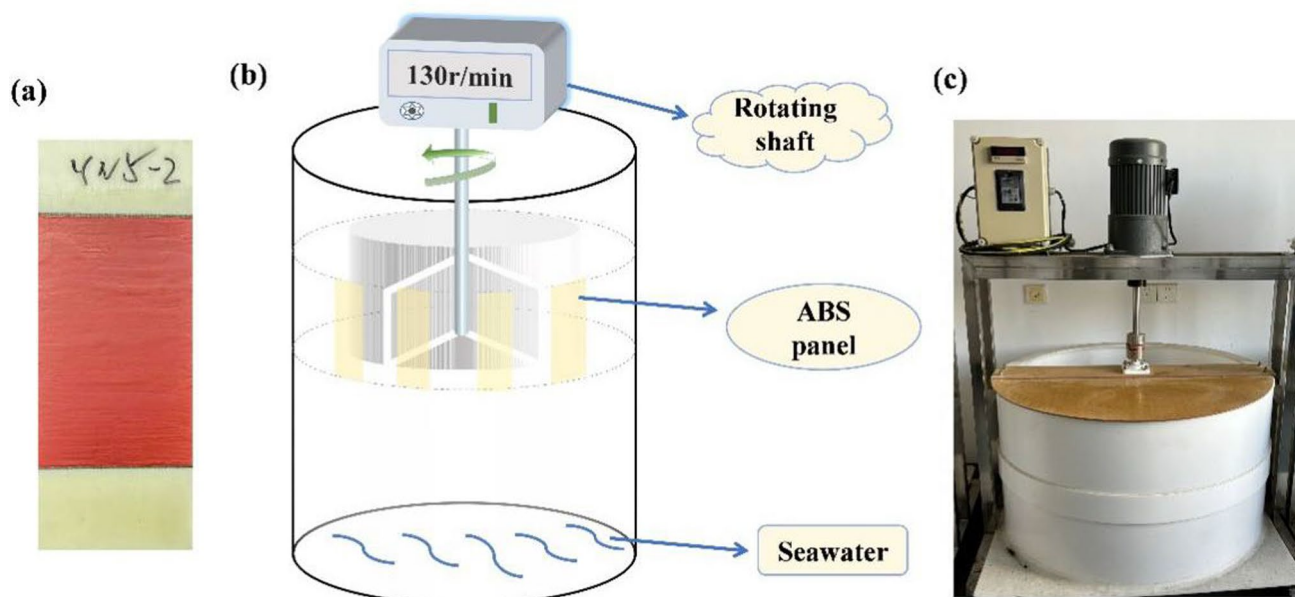


Figure 1 Dynamic padding simulation setup: **a** ABS plate used for dynamic simulation experiments, **b** diagram of the dynamic simulation experiment setup, **c** photograph of the dynamic simulation experimental setup.

of mass rate, surface morphology, and roughness of the ABS plates with different scratching times were measured by contact angle measuring instrument and laser confocal microscope.

Marine environment antifouling test

According to the national standard of the People's Republic of China 'Antifouling Paint Template Shallow Sea Immersion Test Method' (GB 5370-85), the antifouling performance of fluorescent antifouling coatings was determined in the real marine environment [27]. Firstly, the antifouling coating was evenly applied to the sanded PVC board at a thickness of 600 nm and dried naturally in a well-ventilated hood. After ensuring the samples were completely dry. The experimental panels were immersed in seawater at a depth of about 1 m on the artificial raft of Qingdao port [28].

Results and discussion

Quantum chemical calculation of FSCAD

The quantum chemical calculation of FSCAD was carried out by density functional theory DFT-B3 LYP, and the fully optimized molecular geometry, bond length, bond angle, dihedral angle data, frontier orbital parameters, and electron cloud density distribution of the compounds were obtained. Taking compound A as an example, its quantum chemical calculation and electron cloud distribution density are shown in Figs. 2 and 3. The calculation data of the FSCAD are detailed in the supplementary materials (Fig. S9–S14).

From the quantum chemical calculation data of FSCAD, it can be seen that the bond lengths of C–C bonds of the four FSCAD are all around 1.38–1.52 Å, which is shorter than the normal value of 1.54 Å, and the bond lengths of C=O bonds are around 1.21–1.24 Å, which is longer than the normal value of 1.2 Å, indicating that the compounds have conjugate effects. The bond lengths of C–N bonds are all around 1.39–1.47 Å, which is shorter than the normal value of the C–N single bond of 1.52 Å, indicating that partial delocalization occurs within the molecule. The bond angle and dihedral angle data show that there is an array of sp^2 hybrid structures in the compound, and the dihedral angle data indicate that the FSCAD has a good rigid planar structure. The calculations show

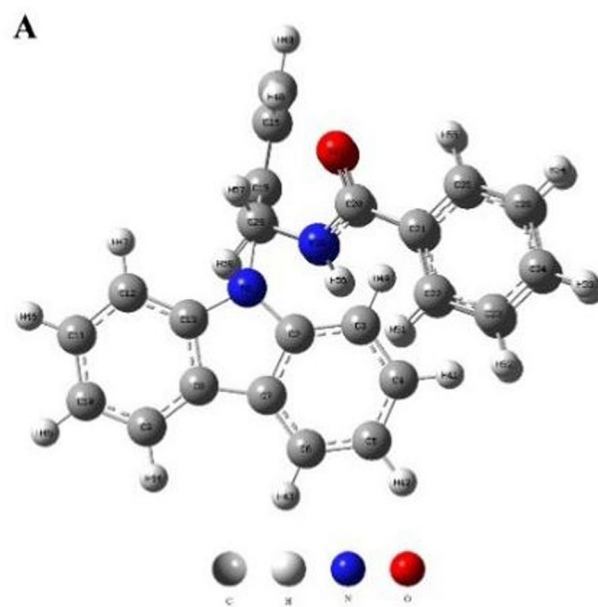


Figure 2 Optimal molecular structure of compound A.

that the compound has good fluorescence, and the dipole moment of compounds A and B is large, indicating that it has good solubility.

Fluorescence property test of FSCAD

Four FSCAD were dissolved and the fluorescence emission spectra were shown in Fig. 4.

The excitation wavelengths of the four FSCAD are 285.0 nm for A, 270.0 nm for B, 285.0 nm for C, and 271.0 nm for D, respectively. The resulting fluorescence intensities are shown in Fig. 4. The order of fluorescence intensity is $C > B > A > D$. The fluorescence intensity of compound C is the highest, exceeding 8000 a.u., and the rest of the compounds are more than 5500 a.u., which could be attributed to the FSCAD contain multiple coplanar benzene ring structures. It is beneficial to improve the π electron conjugation degree and produce strong fluorescence. Therefore, the four FSCAD have strong fluorescence properties, in agreement with quantum chemical predictions.

Antibacterial performance test of FSCAD

The inhibition rate of FSCAD against marine bacteria was determined by the dilution coating plate method [29]. The number of bacteria can be obtained by counting the panel effect diagram of marine bacteria, and the result is shown in Fig. 5.

Figure 3 The electron cloud density distribution of compound A.

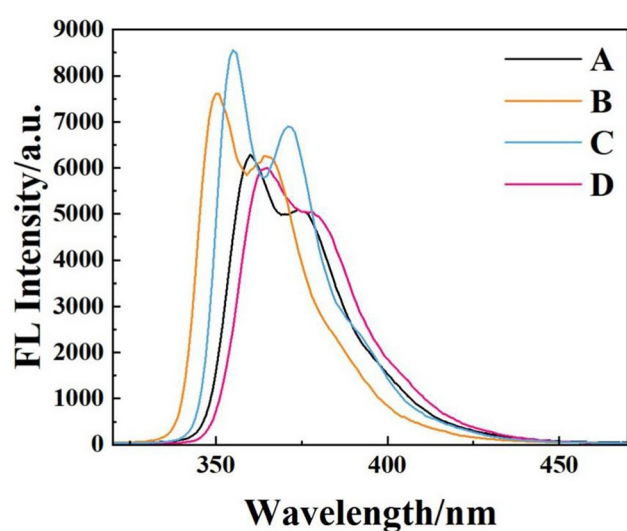
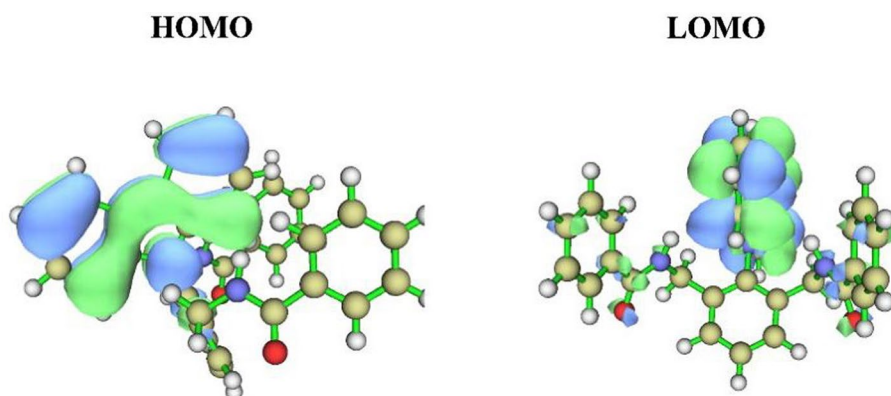
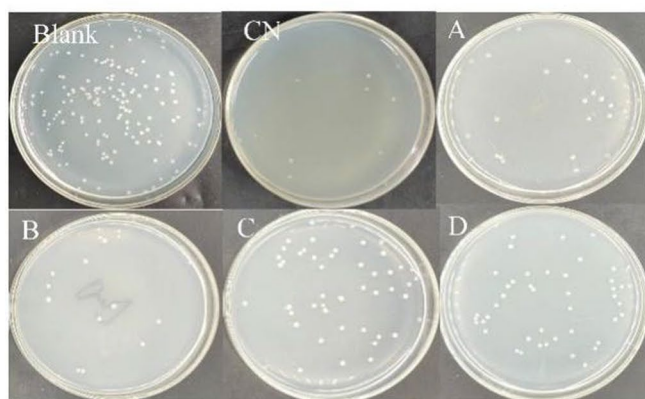


Figure 4 Fluorescence emission spectra of FSCAD.

It can be seen from Fig. 5, that a large number of bacterial colonies appeared in the blank group, while only a small number of colonies grew on the experimental group, indicating that the four FSCAD inhibited bacterial growth and reproduction. Among the experimental groups, compound B exhibited the best antibacterial effect, reaching approximately 90%. This is comparable to the efficacy of chlorothalonil, a broad-spectrum fungicide currently available on the market. Furthermore, the bacteriostatic rates of the other three compounds were all above 60%. This is attributable to the FSCAD contains amide groups, which are biologically active and exert a pronounced inhibitory effect on *Marine Pseudomonas* [25]. And combined with the frontier orbital theory [30], the magnitude of ΔE_{L-H} is indicative of the strength of the molecular reactivity. The smaller the ΔE_{L-H} is, the easier the

(a)



(b)

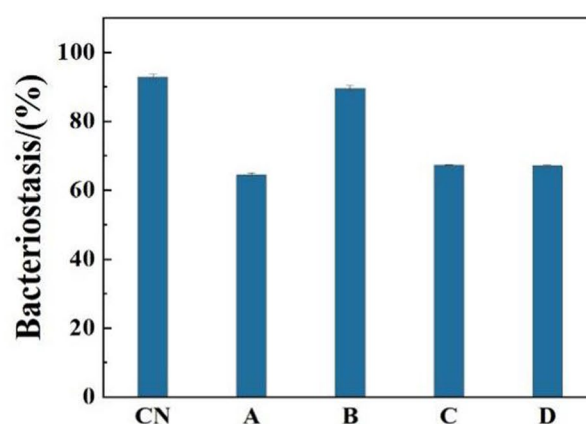


Figure 5 Inhibition of *Pseudomonas aeruginosa* by FSCAD: **a** The growth of *Marine Pseudomonas* **b** inhibition rate.

electron transition is to react with biological macromolecules, thus the better the antibacterial activity is [31, 32]. Because of the smallest ΔE_{L-H} of compound B (Table S6), the bacteriostatic rate was the highest. In summary, the FSCAD exhibits biological activities, which can damage the cell membrane of bacteria, cause abnormal growth of bacteria, and inhibit the reproduction of bacteria [33, 34].

The inhibitory effect of FSCAD on microalgae

The correlation coefficients of the standard working curves of two kinds of microalgae were all above 0.99, indicating that there was a good linear relationship between the absorbance and the concentration of algae at 680 nm [35]. (Fig. S16).

It can be seen from Fig. 6, that the compound C exhibited the highest inhibition rate on *Phaeodactylum tricornutum* and *Chlorella pyrenoidosa* up to 81.9% and 77%, and the inhibition rate of the other three compounds on microalgae was above 75%, which was higher than that of the control group. It has been demonstrated that the Indole derivatives containing amide groups, promote the outflow of Ca^{2+} from microalgae cells, thereby inhibiting the formation of microalgae cell wall, and subsequently inhibiting the growth and reproduction of microalgae [24, 36]. FSCAD possess similar amide groups, enabling their invasion of microalgae cells, damage to organelles, and destruction of the photosynthetic system, thereby achieving the inhibitory effect.

Preparation and characterization of fluorescent antifouling coatings

According to Supplementary Material (Table S1), the synthesized FSCAD were added to the antifouling coatings as antifouling agents. The antifouling properties of FSCAD were compared with the chlorothalonil.

The EDS mapping data of the control group and the experimental group for C, N, O, Na, Mg, Al, Si, Fe, Cu, and Zn were shown in Fig. 7, and evenly distributed and completely overlapped, which proves the good dispersion of the coating.

As can be seen in Fig. 8, the control group exhibited a higher content of Cu elements, while the coating with Cu_2O replaced by Compound A exhibited a lower content, proving that the coating of the experimental group was successfully prepared.

Anti-algae sedimentation test of fluorescent antifouling coatings

In the process of biofouling formation, biofilm formation is the key stage, and diatoms were important participants in biofilm formation [37]. *Nitzschia closterium* and *Phaeodactylum tricornutum* were selected as the tested diatom species, and one green algae *Chlorella pyrenoidosa* was selected as the tested green algae species for analysis in a fluorescent antifouling paint sedimentation test. The number of algae sedimented to the surface of the coating was studied by simulating the natural state and the dark state.

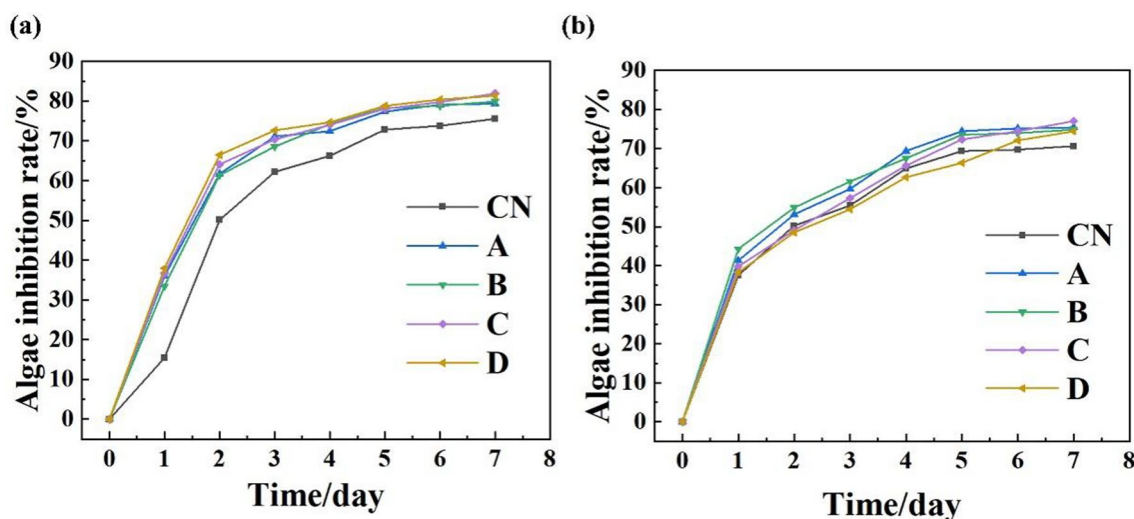


Figure 6 Inhibition rates of microalgae by FSCAD: **a** *Phaeodactylum tricornutum*, **b** *Chlorella pyrenoidosa*.

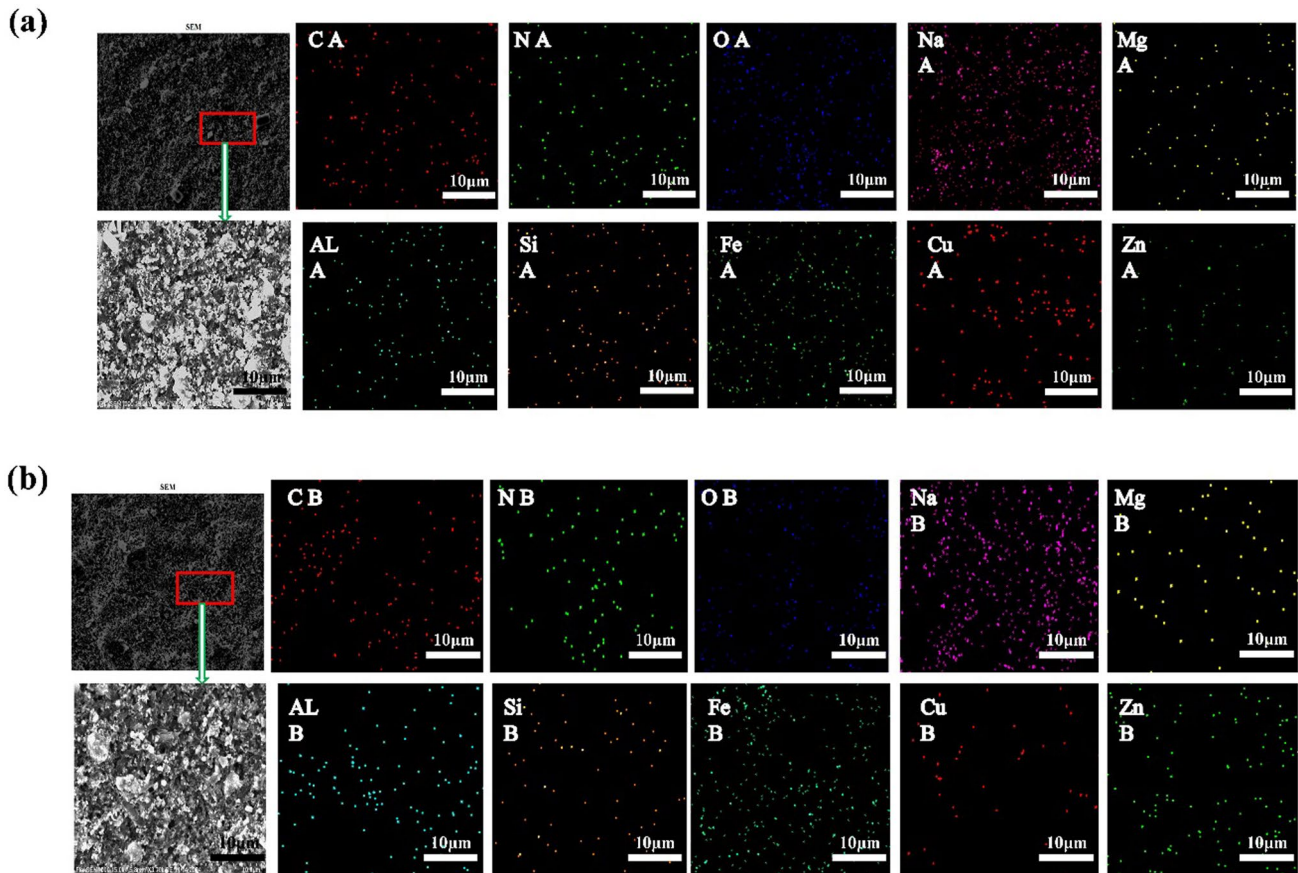


Figure 7 SEM image with EDS analysis: **a** control group T-0, **b** experimental group.

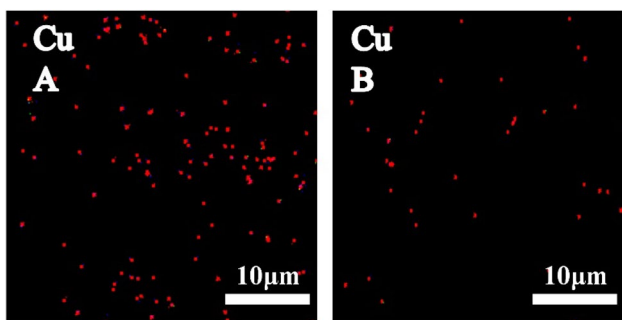


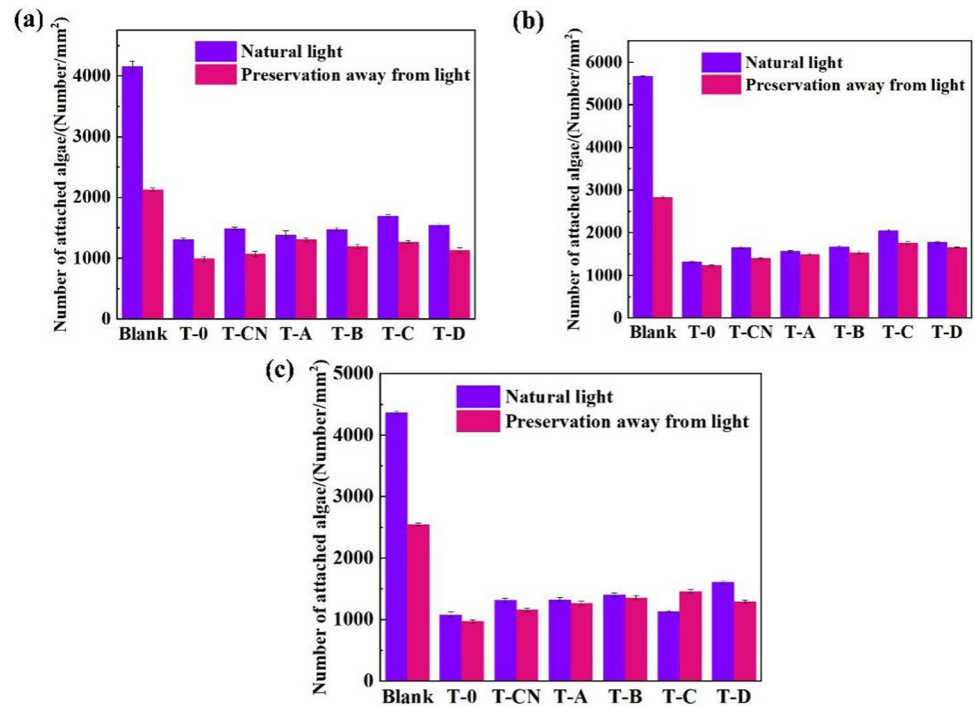
Figure 8 EDS analysis of Cu: **A** control group T-0, **B** experimental group.

As shown in Fig. 9, the four fluorescent antifouling coatings showed excellent anti-algae sedimentation effects on the three microalgae in both states, and the number of sedimented microalgae on the coating was significantly reduced compared with the blank group. In the dark state, the lowest number of sedimented microalgae was observed in T-0. It can be attributed

to the addition of Cu_2O to T-0, which was more toxic to the three microalgae species and therefore exhibited the best resistance to attachment. Compared with the blank group, although the anti-algae sedimentation performance of the coating in the experimental group exhibited a relatively small change, it also demonstrated anti-algae sedimentation performance comparable to that of the control group. This can be attributed to the chemical repellent effect.

Under the natural conditions, T-A exhibited the best inhibitory effect on the two kinds of diatoms, with an inhibition rate exceeding 72.5%. Its anti-algae sedimentation performance was superior to that of the control group T-CN. The inhibitory effects of T-B, T-C and T-D on the two kinds of diatoms were approximately 68.7%, and their anti-algae sedimentation performances were equivalent to that of the control group T-CN. This demonstrates that the coatings in the experimental group have significant anti-algae sedimentation properties against diatoms. The anti-algae sedimentation rate

Figure 9 Number of three microalgae attached to different coating surfaces after seven days in the natural state and the dark state **a** *Nitzschia closterium*, **b** *Phaeodactylum tricornutum*, **c** *Chlorella pyrenoidosa*.



of T-A and T-C against *Chlorella pyrenoidosa* reached 74%, which was better than that of the control group T-CN. The inhibition rate of T-B and T-D against *Chlorella pyrenoidosa* was 67.8%, close to that of the control group T-CN. The results demonstrate that under the natural conditions, the coatings of the experimental group possess excellent anti-algae sedimentation performance.

Compared with the dark state, the coatings in the experimental group exhibited more excellent anti-algae sedimentation performance under natural conditions. According to the study, fluorescence can interfere with the chloroplast spatial structure of microalgae cells, so that the microalgae firstly carry out photosynthesis rather than settlement and attachment, resulting in inhibiting the sedimentation of algae condition [17, 38]. In combination with the anti-algae sedimentation experiment, the experimental group incorporated FSCAD, which emits high-intensity fluorescence upon excitation and effectively inhibits the adhesion of microalgae. Therefore, it is an effective method for coatings to utilize fluorescence repellency to achieve the performance of anti-algae sedimentation, and it has more advantages than chemical repellency.

Effects of FSCAD on the morphology of microalgae cells

The morphology of algae cells in the blank group and the experimental group were shown in Fig. 10. For comparison, the algae cells in the blank group were complete, and the cells in the experimental group showed obvious deformation shrinkage membrane damage, cell debris increased significantly, causing more serious damage. It is inferred that FSCAD promotes the outflow of Ca^{2+} from the algal cells, resulting in damage to its cell wall [33].

Self-polishing performance analysis of fluorescent antifouling coatings

Analysis of abrasion rate and contact angle results based on dynamic simulation experiments

The dynamic simulation experiment simulates the state of the ship and marine equipment being washed by seawater. The side chain structure of the acrylic resin in the coating can hydrolyze with seawater, and the coating is washed and peeled off, resulting in a

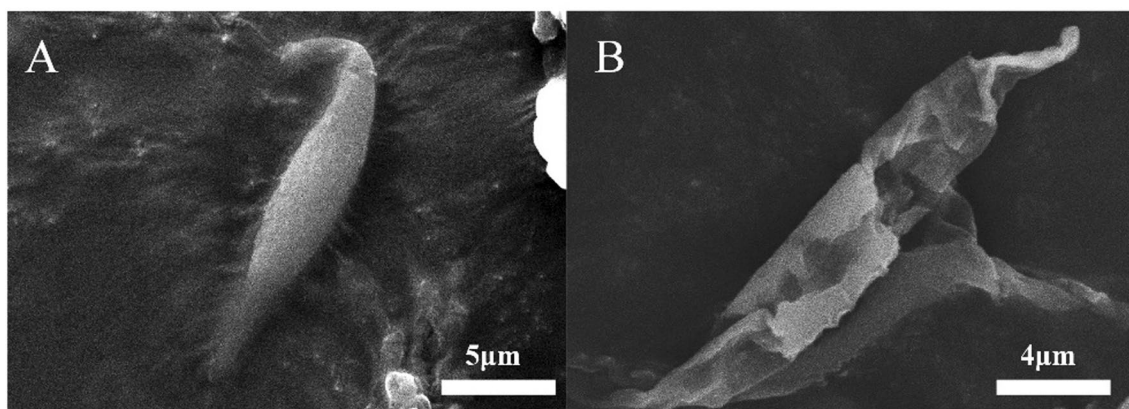


Figure 10 SEM images of *Phaeodactylum tricornutum* cells **A** blank group, **B** experimental group.

self-polishing effect. Therefore, the abrasion rate of the coating is one of the important criteria to evaluate the self-polishing performance of the coating.

As shown in Fig. 11, the abrasion rate of the coating displays a comparable loss pattern, exhibiting high mass loss at the initial stage (about 20 days), which is due to the dissolution of volatile small molecule compounds such as residual solvents in the coating. After 10 days, the mass loss of the coating gradually tends to be stable, indicating that the coating is close to the uniform polishing state. In addition, the contact angle remained above 90° during the dynamic simulation test of 50 days, indicating that the coating exhibited hydrophobic properties, with the relatively small

fluctuation in contact angle of the coating. Therefore, the introduction of FSCAD into the antifouling coating system will not change the contact angle of the coating.

Analysis of dynamic simulated surface morphology change of fluorescent antifouling coatings

The evaluation of the self-polishing performance of the coating is further facilitated by the analysis of the surface morphology through laser confocal microscopy. This approach allows for the direct observation of the morphological changes occurring at the coating surface. Taking T-A as an example, the surface

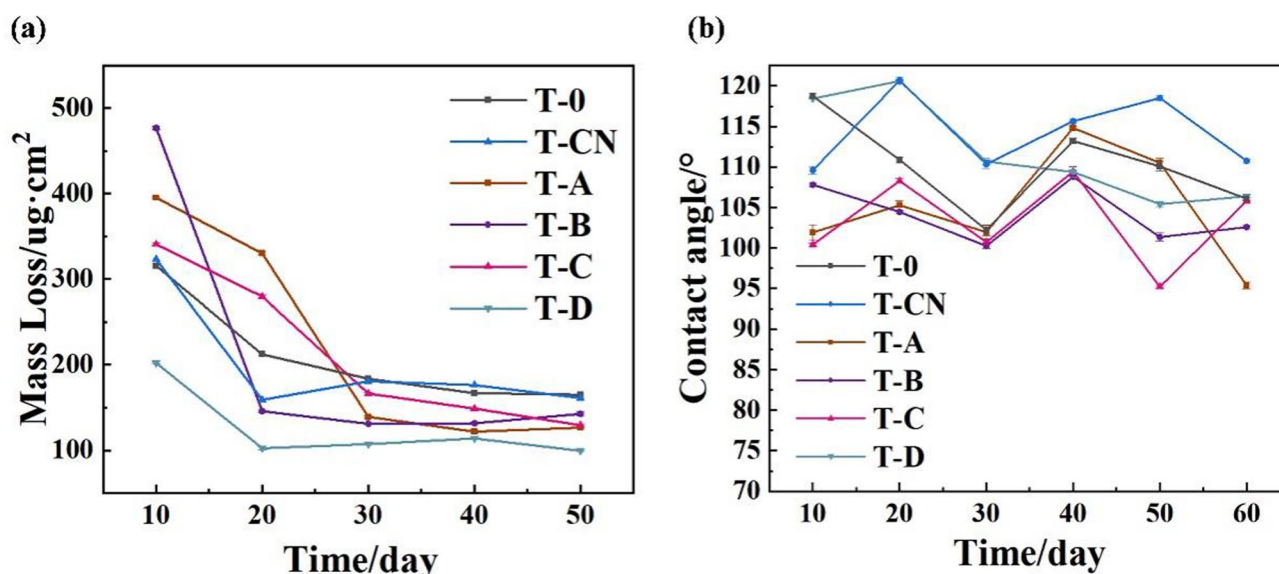
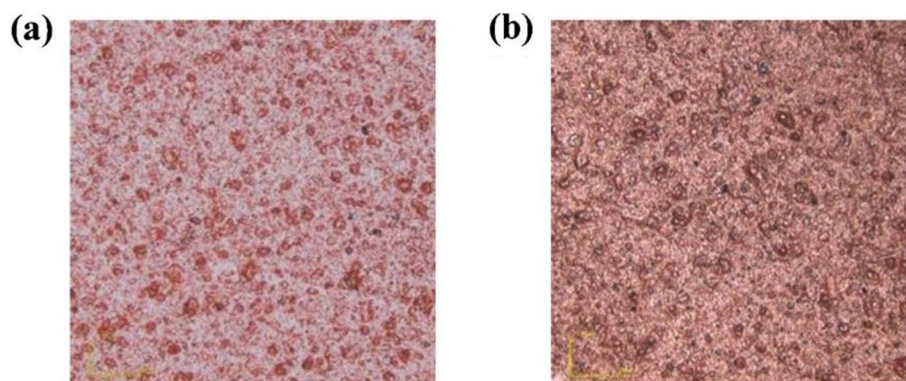


Figure 11 Dynamic simulation experiment of fluorescent antifouling coatings: **a** abrasion rate, **b** contact angle.

Figure 12 Surface morphology of the coating after the dynamic simulation experiment: **a** 0 day, **b** 50 days.



morphology of the coating before and after the experiment is shown in Fig. 12. The surface morphology of all coatings after dynamic simulation experiments is shown in supplementary material Fig. S15.

As can be seen in Fig. 12, after 50 days of the experiment, the surface morphology of the coating barely changed, and there is not exhibited any cracking or detachment, which proves that the coating does not affect the mechanical properties after polishing.

Roughness analysis before and after dynamic simulation experiment

As can be seen from Fig. 13, the coating's roughness demonstrates an overall increasing trend after five cycles of dynamic simulation experiments, with a fluctuation range of 29–41%. This is attributed to the rigid structure containing multiple benzene rings in the fluorescent functional structure, which reduces the flexibility of the coating to a certain extent. The overall fluctuation range of the four fluorescent antifouling coatings is not significant, indicating that the coating possesses relatively good smoothness which helps to reduce the adhesion of fouling organisms.

Analysis of marine antifouling results of fluorescent antifouling coatings

According to the bacteriostatic experiment and the microalgae growth inhibition experiment, the four FSCAD have exhibited inhibitory biological activity. When the four FSCAD were added to the marine antifouling paint as an antifouling agent, the results of the marine environment antifouling experiment are shown in Fig. 14.

The antifouling performance of fluorescent coatings in the real marine environment was tested in the Qingdao

Sea area of Shandong Province (Fig. 14). After immersion in the Qingdao Sea area for 3 months, the surface of the blank panel was covered with biofouling, such as algae, animal larvae, and mussels. On the control groups T-0 and T-CN, algae and a large amount of biological mucosa were adhered. In contrast, the experimental groups T-A, T-B, T-C, and T-D adhered relatively few biofouling coatings, indicating that FSCAD can effectively inhibit the adhesion of marine organisms.

After immersion in the Qingdao Sea area for 6 months, the surface of the blank panel was covered with big biofouling. The surface of T-A exhibited minimal biological fouling, demonstrating the antifouling performance is the best, which is better than T-CN and T-0. The surface of the other three coatings only adhered to a small part of biological mucosa, algae, and sediment, with no large antifouling biological attachment, showing good antifouling performance better than the control group T-CN and close to T-0.

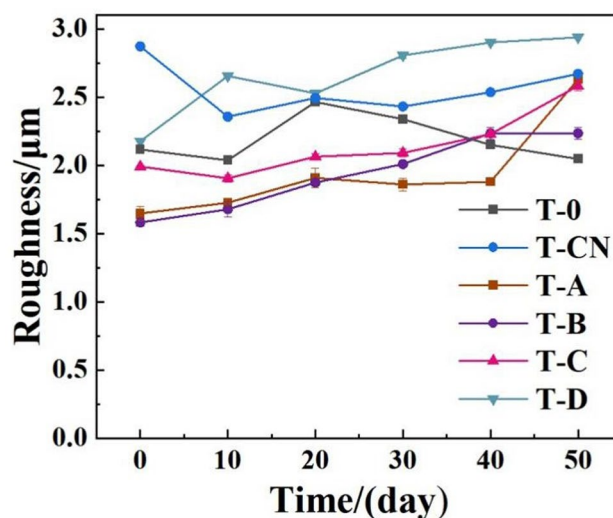


Figure 13 Roughness change of fluorescent antifouling coatings after dynamic simulation experiments.

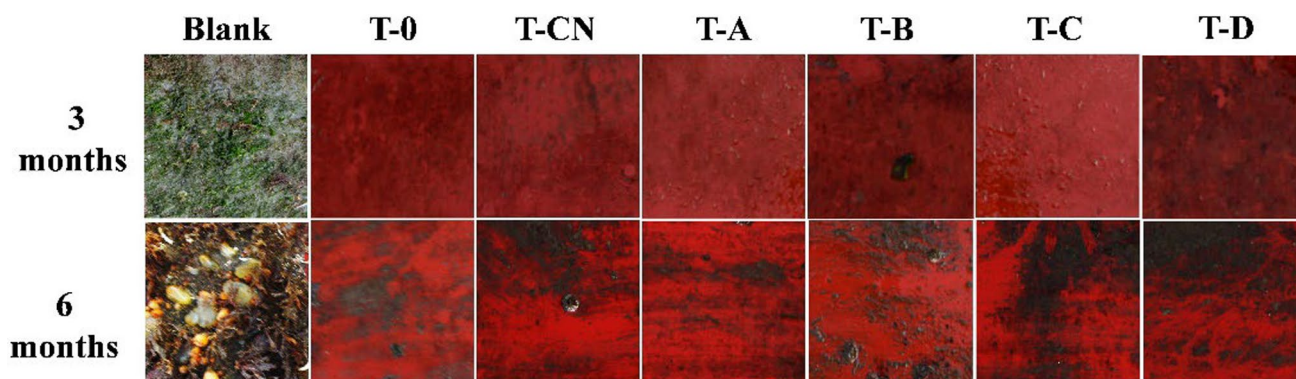


Figure 14 Image of coating surface immersed in sea for 3 and 6 months.

The marine antifouling results indicate that when the FSCAD was partially substituted for Cu_2O as an antifouling agent in the coating, the coating still exhibited good antifouling performance. This study provides a theoretical basis for fluorescent amides as antifouling agents.

Antifouling mechanism of the fluorescent antifouling coatings

The antifouling mechanism of coatings is shown in Fig. 15. Four kinds of FSCAD were substituted for Cu_2O as antifouling agents in the antifouling coating

system. The presence of amide groups in FSCAD results in high chemical reactivity. By controlling the outflow of Ca^{2+} from microalgae cells, these derivatives can disrupt the formation of microalgae cell walls, subsequently inhibiting the growth and reproduction of microalgae [33, 35, 39]. More importantly, the high-intensity fluorescence emitted by the coating after excitation can change the spatial structure of chloroplasts in algal cells so that microalgae prioritize photosynthesis over sedimentation and attachment [17, 18, 40]. Lastly, due to the addition of acrylic resin, which can ion exchange between Zn^{2+} in the side chain and Na^+ in the seawater, achieving the self-renewal of

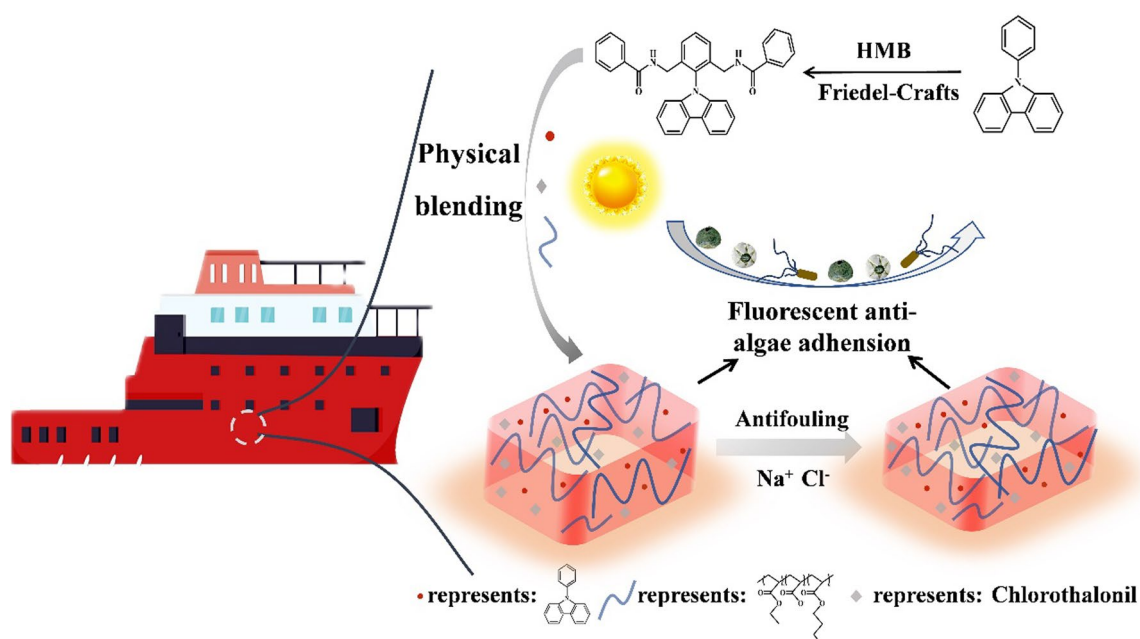


Figure 15 Antifouling mechanism study of FSCAD.

the coating surface [41, 42]. Additionally, hydrophobic groups are generated on the coating surface, rendering the coating hydrophobic. This allows antifouling organisms attached to the coating to be peeled off from the resin surface by seawater, effectively reducing the adhesion of the fouling organisms.

In summary, the fluorescent antifouling coatings combine the chemical and fluorescence properties of FSCAD and the self-polishing properties of the acrylic resin, which can significantly reduce the use of Cu_2O in the current self-polishing antifouling coatings while maintaining good antifouling properties.

Conclusion

Four FSCAD were successfully synthesized through Friedel–Crafts alkylation reaction. Among them, compound B showed the best bacterial inhibition effect, which was close to 90%, and the bacterial inhibition rate of the remaining three compounds was higher than 60%, which were incorporated into coatings as antifouling agents to replace Cu_2O . This allowed the coating to exhibit significantly improved algae inhibition and anti-algae sedimentation adhesion. The coatings blended with four FSCAD exhibited better anti-microalgae sedimentation effects under natural conditions. In particular, compound A demonstrates the highest biological activity and the anti-algae sedimentation rate can reach 79.5%. Observing the morphology of microalgae, the microalgae cells in the blank group were complete, and the cells in the experimental group showed obvious deformation shrinkage membrane damages. After 6 months of marine environment antifouling experiments, the T-A coating exhibited the best antifouling performance, superior to the antifouling coating T-CN with chlorothalonil. There was no obvious adhesion of large fouling organisms on the surface of the other three coatings, and the antifouling performance was close to that of T-CN.

The new self-polishing antifouling coatings prepared in this study can significantly reduce the use of Cu_2O in the current self-polishing antifouling coatings while maintaining good antifouling performance, which provides a new approach for the development of environmentally friendly marine antifouling coatings with both fluorescence and chemical activity.

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Author contributions

YW and WD involved in data curation, formal analysis, investigation and writing—original draft; XY took part in visualization; XL involved in funding acquisition, resources, supervision and writing—review & editing; LY involved in conceptualization and project administration.

Data and code availability

All data generated or analyzed during this study are included in this published article and its supplementary information files.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical approval Not applicable.

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